The Ontology of a Product-Based Circular Economy Implementation Framework for the Manufacturing Industry

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Chapter 19
The Ontology of a Product-Based Circular Economy Implementation Framework for the Manufacturing Industry

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ABSTRACT

The world economy has been built around a linear economy, that does not account for the impact on environment or operations of the industry. The concept of circular economy is being adopted by countries across the world to address this. In this chapter, a circular part repository business model and product-based circular economy implementation framework are presented to evaluate the environmental impact of the product manufacturing industry through the quantification of circular economy quotients. The approach to implementation of the framework onto a product lifecycle management platform is also explained. This would serve as the basis for building a circular part repository that could form the central core of existing eco-industrial parks thereby structuring the reuse of recycled material and minimising new raw material intake.

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1. INTRODUCTION

The linear economy, grounded on a 'take-make-dispose' framework, systematically extracts resources, manufactures products, and discards them post-use, engendering a cycle of waste and inefficiency Geissdorfer et. al (2017). This paradigm is increasingly criticized for its unsustainable use of finite resources, leading to deleterious environmental impacts, such as pollution and depletion of natural ecosystems Kircherr, Reike & Hekkert (2017) . By reimagining economic activity to detach growth from finite resource consumption, the circular economy seeks to establish a regenerative system that retains and recovers value from products and materials, thereby mitigating the linear economy's drawbacks and paving the way for sustainable industrial practices Stahel (2016).

The product manufacturing industry accounts about 16% of global GDP (Stromquist, 2019). It is pivotal, underpinning economies worldwide, by generating significant employment, fostering innovation, and driving export revenues Wang et al. (2023). Globally, legislative bodies are intensifying efforts to curtail the environmental impact of the product manufacturing industry. Specific targets include increasing recycling rates to 50% by 2030 in the United States and 64% for packaging waste in the European Union by 2025 Tumu, Vorst, Curtzwilier (2023).

The following example shows how the automotive industry, with its wide network of manufacturing and ancillary units, is a case in point. Directive 2000/53/EC of the European Parliament and of the Council on end-of-life vehicles (ELV) states that from January 2015, reuse and recovery for vehicles shall be increased to 95% of average weight per vehicle and year while reuse and recycling shall be a minimum of 85%. Japan and EU have very good reuse, recovery and recycling levels with reuse and recovery standing at above 80% and reuse, recovery and recycling at more than 95% Bhari, Yano & Sakai (2021). However, some established economies like Australia and emerging world economies like India and Brazil are not on the same level. Australia does not have a formal ELV legislation despite a huge number of deregistered vehicles and hence the recycling efficiency, while more than 95% for iron (Fe), is much lower for aluminium (Al)/zinc (Zn) and copper (Cu) being around 40% and 4% respectively. Overall, the recycling and recovery efficiency in Australia was around 72% (Soo, 2017). Technology emerges as a key enabler in overcoming challenges towards a more circular economy as it offers the potential for more circular resource flows and waste reduction Bocken, Nuzum, Weissbrod (2021).

The concept of a circular economy (CE) has gained significant traction in recent years as a sustainable alternative to the traditional linear economic model of 'take-make-dispose' Blomsma and Brennan (2017). The advent of the circular economy as a paradigm for sustainable development has garnered significant attention, particularly within the manufacturing sector Kirchherr, Reike and Hekkert (2017). In the context of the manufacturing industry, this shift towards circularity is not just a trend but a necessary evolution to address the pressing environmental challenges and resource scarcity issues facing our world today Geissdorfer et. al (2017), as well as the complexities of global value chains that significantly impact environmental outcomes Campos and Rodi (2024). The CE model emphasizes the retention of value within products, materials, and resources, aiming to create a regenerative system that minimizes waste and maximizes resource efficiency Camilleri (2020).

This involves a thorough exploration of current practices, challenges, and opportunities Centobelli et al (2020). By first defining the industry's current standing in relation to CE by developing a Circular Economy Quotient (CEQ) metric and then detailing a Product-Based Circular Economy Implementation (PCEI) framework necessary to achieve compliance, the research provides a structured pathway for transforming from linear to circular processes. Moreover, the implementation of the framework through

Product Lifecycle Management (PLM) systems, as discussed in this paper, plays a crucial role in building a comprehensive CE implementation model. PLM enables the tracking and management of a product's entire lifecycle, from initial design through end-of-life, in alignment with CE principles. This approach facilitates the efficient use of resources, optimizes product reuse and recycling, and ensures compliance with environmental regulations. This highlights the necessity of not only adopting new technologies but also ensuring that they align with the overall goals of sustainability Zheng et al. (2018). This also emphasizes that while leveraging technologies, it is important to focus on sustainable design, product use maximization, and the development of new business models Bocken, Nuzum, Weissbrod (2021).

This chapter introduces an innovative PCEI framework, which effectively bridges the gap between theoretical constructs and practical applicability, a limitation often noted in existing CE literature. This framework is based on recent advancements in Industry 4.0 and developments in the manufacturing sector Khan, Ahmad & Majava (2021). The rationale for this study emerges from the pressing need to mitigate the environmental impacts of manufacturing processes while maintaining economic viability and promoting societal well-being, aligning with the closed-loop and product service systems Camilleri (2019). The aim is to address the gap in the literature concerning the practical implementation of CE principles in the manufacturing sector, particularly focusing on the integration of technology as a facilitator for CE (Bocken, Nuzum, Weissbrod 2021).

This chapter begins an extensive review of existing literature to establish the theoretical underpinnings of the study. Subsequent sections delve into the methodology adopted, providing a detailed account of the approach and techniques employed in the research. Following this, the introduction of the Circular Part Repository (CPR) business model is discussed, which encompasses the introduction of the concepts of Environmental Impact Quotient (EIQ) and Environmental Costs. The discussion extends to the computation of the CEQ, offering insights into the quantification of circularity within product manufacturing.

2. CIRCULAR ECONOMY

Researchers (Delagrade & Wassenhove, 2017) have attempted to define circular economy (Ellen Macarthur Foundation 2015, Hass et al 2015, Masi et al 2018, Geissdorfer et. al (2017) by focusing on policy (Murphy and Gouldson 2000, Zihun and Nailing 2007, Sakai et. al 2016, Dalhammar 2016) renewable energy (Tu et al 2011, Chang et al 2011, Feng et al 2012, Reh 2013, Ampelli, Perathoner & Centi 2015) and other related issues.

Circular economy is focussed on cyclical thinking rather than an open-ended conceptual approach (Bocken et al 2016, Wubbeke and Heroth 2014). CE also focuses on minimizing the consumption of virgin materials and energy (Wubbeke and Heroth (2014). CE is fast emerging as an economic strategy rather than a purely environmental approach (Yuan, Bi & Moriguchi (2006). This means that industrial structure and industrial policy reform must be adjusted to promote new technology development to reach a solution by changing the waste recycling focus Tu et al. (2011).The linear economy is not sustainable (McDonough & Braugart 2002).as the world runs on limited resources (Hanson & Nicholls, 2020). Circular Economy, can be defined as a concept whose implementation entails reducing the consumption of raw materials, designing products in such a manner that they can easily be taken apart and reused, prolonging the lifespan of products (van Buren et al., 2016). As an outcome, waste is reduced by closing loops in industrial ecosystems by replacing virgin production with the availability of the renewed (Stahel, 2016).

This process minimises material flow, energy-flow and environment deterioration without restricting economic growth (Stahel, 1982) and creates, by design, a restorative or regenerative industrial economy, eliminates the concept of end-of-life (Geng and Doberstein 2010, Ellen MacArthur Foundation 2015,) through multiple non-linear phases (Yuan, Bi & Moriguchi, 2006). Circular economy fosters sustainable production emphasizing operational efficiencies and waste reduction Camilleri (2019). It can also be characterised as the design and business model strategies that are slowing, closing and narrowing resource loops (Bocken et al (2016).

2.1 Circular Business Models

Businesses cannot be completely altruistic and will have to be economically viable while being environmentally sustainable Achmad (2022). As described by Pieroni, McAloone & Pigosso (2019), business models need to be developed that create capture economic value . In circular economy, business model innovation plays an important role to fundamentally change the way of doing business to go beyond prevalent sustainability approaches that focus on efficiency, productivity, and 'greening' the supply chain Bakker et al (2014).

A morphological analysis of 26 current Circular Economy Business Models (CEBM) from both academic and practitioner literature was carried out Ludeke-Freund, Gold & Bocken (2018) leading to identification of six major patterns to classify CEBMs as (i) repair and maintenance (ii) reuse and redistribution (iii) refurbishment and remanufacturing (iv) recycling (v) cascading and repurposing and (vi) organic feedstock which are mapped against the four core principles of business models namely (i) value proposition (ii) value delivery (iii) value creation and (iv) value capture. It was found that there is a shortage of tools for extending implementation of circular business models for environmental assessment and guidance Goffetti et al. (2022). This calls for a proper framework with measurement tools to be embedded into the circular business model framework itself.

2.2 CE Frameworks

The complexity in the definition of CE has led to the development of different frameworks to better support the practical implementation of CE. CE frameworks are usually based on the business models that they are meant to cater to. The categorisation of a few frameworks based on the business models is discussed herein.

Sustainable Business Model Innovation (SBMI) framework outlines processes and solutions that can help organisations embed sustainability in their business model by taking a value-network approach by altering the organization's value delivery and/or capture and/or propositions. However, neither paradigm provides systematic (top-down or bottom-up) direction on how a company might find, analyse and execute CE opportunities by strategically connecting business model innovation with product design requirements. This disconnect was identified and a CE product and business model strategy framework that combines future visioning and goal setting to guide circular product design was described while also considering CE business model strategies (Bocken et al, 2016). This framework, on the other hand, does not provide step-by-step instructions on how organisations should build an overarching vision on CE principles and effective action plans to develop CE and to address the essential business and product design adjustments to actualize the vision.

Closed-Loop Systems (CLS) frameworks have been developed with the aim of closing material loops. They are also consistent with the CE principles in this regard Lifset & Graedel (2002). Resource conservative manufacturing, for example, views resource conservation and closed-loop systems as important parts of product design and development Rashid et al (2013). Its implementation, however, is difficult since it necessitates significant changes in business models, product design and supply chain architecture.

Selling services rather than things is a fundamental business strategy in the product-service systems (PSS) frameworks a concept created to support a change to a more sustainable economy Camilleri (2019). PSS implementation is now widely regarded as one of the most effective tools for promoting CE (Bocken et al 2016, Tukker 2015, Bakker et al. 2014) . Product-oriented, use-oriented and result-oriented business models are the three types of PSS (Tukker 2004). Successful PSS deployment requires a relationship between strategic and operational decisions (Reim, Parida and Ortqvist 2015). The PSS strategy, according to Baines et al. (2007), must be adopted at the system level since it necessitates changes in organisational structure and early consumer interaction.

Cradle-2-Cradle design (Braungart, McDonough & Bollinger (2007) is one of the frameworks in the category of Sustainable Product Design (SPD), with the goal of assisting designers in the production of eco-effective goods and industrial processes based on three natural principles: waste equals food, utilise sun energy and appreciate diversity McDonough & Braugart (2002). This is accomplished by promoting optimal material flow within technological and biological cycles (Braungart, McDonough & Bollinger, 2007). Biodegradable materials that can be safely returned to the environment to feed biological processes, or technical materials with the potential to be safely reused in CLS, should be utilised in C2C goods. A framework was established by Baumeister et al. (2014) that included step-by-step directions for scoping, discovering, developing and assessing biomimicry thinking in product design based on the concept of biomimicry developed by (Benyus (2002). However, as Volstad & Boks (2012) point out, biomimicry should not be applied without first determining whether nature genuinely has the best solution for a given situation. Similarly, it was pointed out that simply mimicking shapes and processes using biomimicry does not always result in more sustainable products de Pauw et al (2014). As a result, biomimicry can only attain its full potential when applied holistically and at the system level (Montana-Hoyes & Fiorentino, 2014).

Having looked at each of these frameworks, it is evident that they only meet a small number of CE requirements while having their own limitations. All these different frameworks strive to address different parts of the CE, depending on the industry, product type or stage of product development in which they are used. Clearly, based on the academic and practitioner literature there is very little support in this field for the implementation of CE in the product manufacturing domain. There is currently a dearth of comprehensive frameworks for the CE idea despite its widespread use and adoption (Masi et al, 2018). This need requires to be catered to move CE forward in the bottom-up systemic approach.

2.3 CE Implementation and Measurement Tools

As discussed in the above section, there are lots of CE frameworks. CE measurement tools and implementation is not different from the CE frameworks, but it has a measurement tool embedded into it and the nature of such a framework is to focus on not just the implementation of a business model it is based on, but also the measurement of a particular aspect of circularity of a system Garza-Reyes (2018). The Ellen MacArthur Foundation has developed the 'Circulytics' tool based on the Material Circularity Index (MCI) framework for the measurement and quantification of circularity of an entire organization and its processes. This tool calculates the minimization and maximisation of a linear and circular approach. However, due to the complexity of the framework, practical adaptation and application for SMEs may be difficult. On the other hand, a tool was created that evaluates circular production systems in the food and chemical industries by combining life cycle analysis with an environmental input-output analysis Genovese et al (2017). To assess the impact of industrial symbiosis on CE, (Wen & Meng, 2015) proposed a Resource Productivity Indicator. The Circular Economy Index (CEI) proposed by (Di Maio & Rem, 2015) measures the ratio of actual recycled products to what was received for recycling. Yet another framework and measurement tool were proposed by Elia, Gnoni & Tornese (2017) that could be used at all levels, including micro, meso, and macro and that outlines the requirements to measure, processes to monitor and actions to take. This tool does not fully encompass all necessary requirements to be measured, as well as the processes to be monitored and actions to be taken.

The Reuse Potential Indicator (RPI) was proposed by Park and Chertow (2014), which serves as a decision-making tool in recycling processes by identifying characteristics of materials in comparison to resource and waste. For the Chinese chemical sector, a five-category index system was suggested based on economic development, ecological efficiency, pollution reduction, resource exploitation, and development potential (Li & Su 2012). To evaluate the implementation of CE in water recreation parks, an indicator was proposed that included cost-effectiveness, eco-costs, and market value (Scheepens, Vogtlander & Brezet 2016). Though there are a lot of tools that measure the circularity of a product in various aspects, there still isn't adequate support, on the measurement front, for the product manufacturing industry. This lacuna has been attempted to be addressed by Garza-Reyes (2018) through the development of a measurement toolkit for manufacturing SMEs. The Circularity Measurement Toolkit (CMT) was presented therein which enables the assessment of the degree of circularity in manufacturing SMEs. A '9-level' circularity index was proposed which appears to be a at a higher systemic level covering a wider spectrum of CE challenges. However, this seems to rely heavily on a questionnaire-based approach to assess an industry's performance. This does not give scope for the industry itself to quantify the circularity of the product. In order to bridge this gap, it is necessary to develop a bottom-up, product level, circularity measurement tool integrated with a framework built on a sustainable business model (Su et al 2013).

CE Definition: *Circular economy deals with the implementation of an approach to ensure maximum value is extracted from a product in use through rework, recycle and recovery till the end of its service life while minimizing the energy expended in the process.*

Based on the above definition, there is a need to define a sustainable circular economy business model and develop a framework and scorecard that can help a product manufacturing industry quantify its adoption of CE. In this work, a quantitative scorecard is proposed to define the environmental cost and provide an entry point to trace back to the design of the product itself thereby making it more CE compliant. The framework proposed can be used to assess the circularity of products and help a manufacturing industry to evaluate its level of compliance to CE.

3. METHODOLOGY

The Toulmin method (Toulmin, 1958) is adopted herein to build the epistemological basis for a CE solution. A product-based circular economy implementation (PCEI) framework is presented. The Toulmin method outlines two sets of triads with three components to each triad. The first triad makes a *claim* (C) that is grounded on *data* (D) and connected by a logic or reasoning denoted as *warrant* (W).

The second triad adds three components namely *qualifiers* (Q) to the claim, *backing* (B) to the warrant and a *rebuttal* (R) to the claim. The claim is a proposition or lemma which is grounded by observations made or data collected from sources. The data, D, points to the need for definition of the claim, C. The reasoning or rationale behind the proposition is presented in warrant, W and backed with support information, B. The limitations in the applicability of the proposition are presented in qualifiers, Q and any counter proposition described in rebuttal, R. These 6 components put together form the basis of the Toulmin analysis that has been used herein to critically analyze the domain of circular economy and its implementation in industry. The application of the Toulmin method to the problem of circular economy implementation is presented in figure

The individual elements are below.

Claim (C):

Lemma 1: Implementation of circular economy in a manufacturing industry for product P_i (where 'i' indicates number of products in the portfolio) requires computation of $\sum \ddot{Q}_{nR}$ where $\ddot{Q}$ refers to the Environmental Impact Quotient (EIQ) and nR refers to the number of R imperatives applicable to P_i

Data (D):

Industry classification† = **11** sectors; **24** industry groups; **69** industries; **158** sub-industries

Number of companies worldwide‡ = 213.65 million (as of 2020)

Countries implementing CE§ = **21** countries out of **195** worldwide

Industries implementing CE¶ = **9%**

†Global Industry Classification Standard (GICS ®) Methodology, January 2020

‡http://statista.com; §World Economic Forum 2020; ¶https://ec.europa.eu/eurostat

Warrant (W):

Lemma 2: *Only China and the European Union have extensively embraced CE. Despite proposing several CE frameworks, gaps still exist in arriving at a comprehensive bottom-up approach to quantifying environmental impact of product manufacturing.*

Backing (B):

Number of research articles reviewed = 471

Literature survey quantified into tranches = 15 different sub-topics

Published articles on CE quantifiers = 10

Qualifiers (Q):

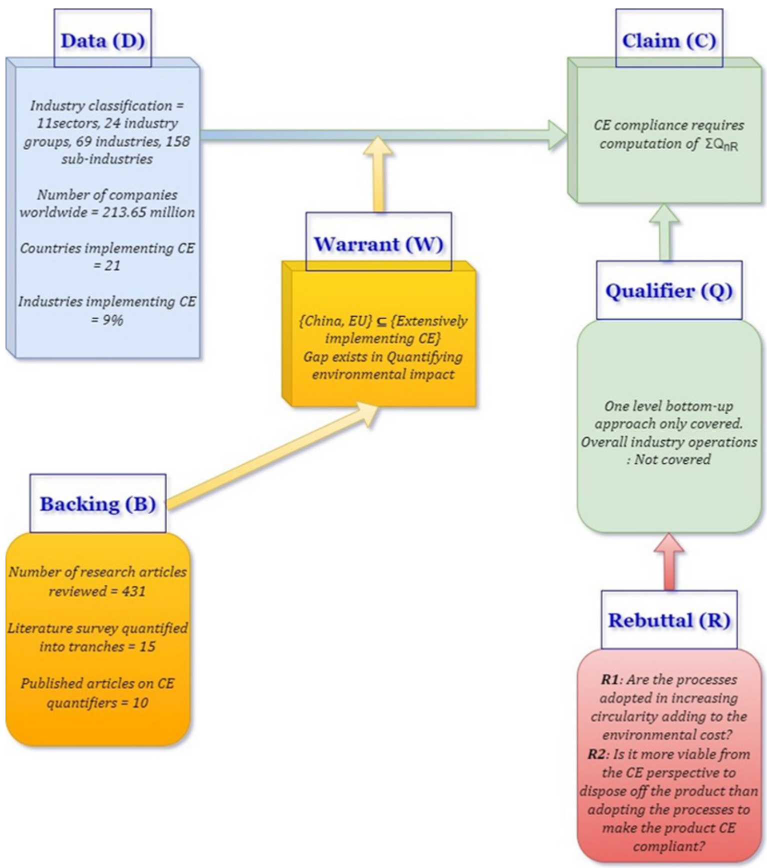
Lemma 3: *The domain of CE is wide and all-encompassing. Arriving at one comprehensive quantitative definition is too complicated and hence not part of Lemma 1. A bottom-up approach from a product and its impact and covering the entire portfolio is required.*

Rebuttal (R):

R1: *Are the processes adopted in increasing circularity adding to environmental cost?*

R2: *Is it viable to dispose product than adopt processes to make it CE compliant?*

Figure 1.Toulmin Approach to CE Claim Definition



The claim is that a product or industry can be assessed for its compliance to circular economy through the computation of its 'environmental impact quotient'. This is based on the grounds or data of the number of industry sectors and companies worldwide, countries implementing CE and industries implementing CE. The reasoning (presented in warrant W) is that gaps still exist in quantifying the environmental impact especially in product-based industries. This is backed by Two major rebuttals (R1 ad R2) could be raised herein to counter the claim C. Rebuttal R1: The processes adopted in increasing circularity of a product themselves involve an environmental cost. Would it be meaningful to do so yet claim increased circularity? This question is addressed in this chapter by the framework through the computation of a circular economy quotient that would quantify the impact of the processes adopted. Rebuttal R2: The second rebuttal questions the very need for making a product CE complaint rather than dispose the product directly. This again can only be confirmed through the computation of the circular economy quotient for direct disposal versus processing for circular economy compliance.

4. CIRCULAR PART REPOSITORY (CPR) BUSINESS MODEL

Industrial symbiosis is gaining worldwide implementation (Yu & Dijkema, 2014) where even the waste from an industry can be utilised by another as resource (Chertow, 2000). Collaborative co-existence is in three stages. In the first stage, there is a random resource exchange without conscious effort of flow of material and energy back into trade rather than into waste. Stage two is a structured exchange of resources and energy among the participating industries through established networks with mutual economic benefits. A maturing of this stage leads to establishment of an institutional framework that links the industries involved and penetrates the activities of collaborative use of material and energy (Chertow & Ehrenfeld, 2012).

Efforts have been made to quantify the contribution made to circular economy by industries within a specific region, termed as eco-industrial parks. While trying to map substance flow, the complexity of the problem has also come to the fore given the issue of data availability, interaction involved between industries and the 'exit-entry' requirements of recycled material between the industries (Wen & Meng, 2015). These parks in China optimise the use of resources (raw material and energy) mostly among material processing industries for them to effectively utilise the waste generated by one industry as resource for the other (Zhang et. al 2010, Dong & Ren, 2015).

Figure 2. CPR Business Model

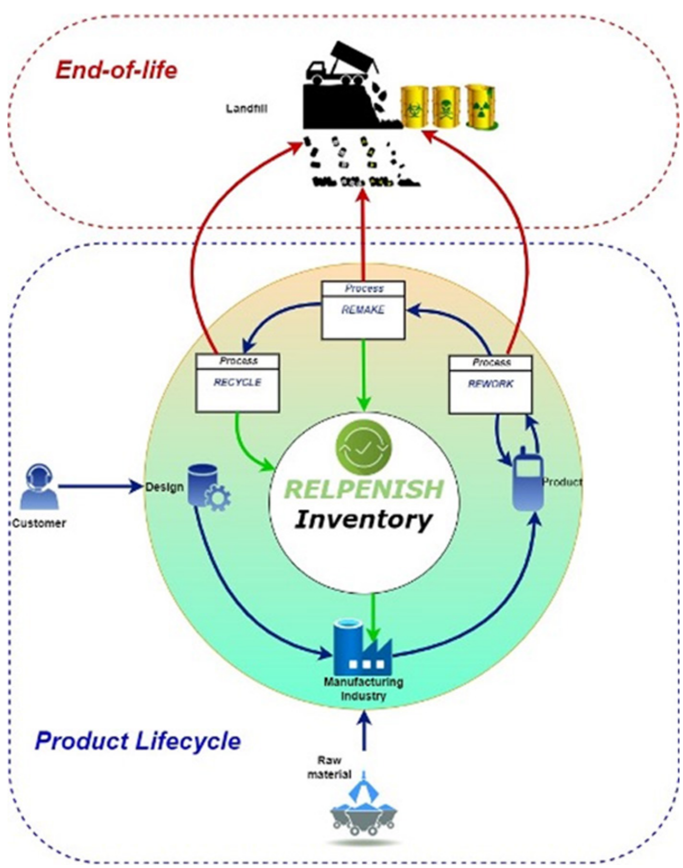


Figure 2 illustrates the circular part repository business model with the core element of 'replenish' as the common part inventory. This model can be used at an individual industry level or at the level of the product manufacturing eco-industrial park. In either case, the central 'replenish' repository is the key inventory that product manufacturing industries must strive to augment both at individual and collective levels.

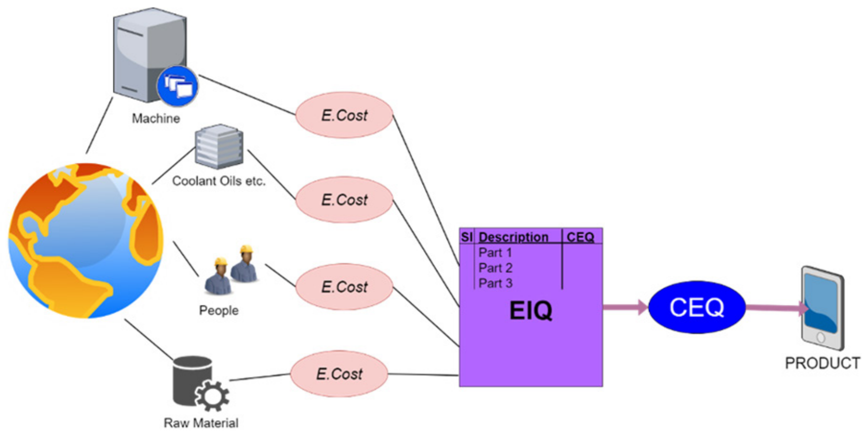
In this model remade parts go into the replenish inventory that can be used as parts for other products. This minimises the intake of raw materials in another dimension thus adding to the circular manufacturing inventory. Those parts that cannot be remade will have to be put through a recycle process that could also produce some parts that go into the replenish inventory. Each of these processes namely rework, remake and recycle invariably produce some scrap that go into the reject inventory list for that industry.

Product lifecycle management (PLM) systems are designed to maintain all data related to a product and its parts in terms of 3D models, documents, workflows etc. The benefit of implementing a PLM system is that all the product data of the industry, sourced from across all its operating units that may be at different geographical locations, are maintained in a central data repository accessible to all the operating units. This helps the units and the industry at large to track every part throughout its lifecycle. Extending this to the concept of the central part repository can similarly help the industry, or the product manufacturing eco-industrial park, to track and maintain inventory of all replenished parts that is available to all the operating units or constituent industries. Tagging of these parts, monitoring their product lifecycle and manufacturing history with the help of the PLM system, can help these symbiotic industrial parks to easily source and manage the central part repository. This circular part repository (CPR) business model founded on a PLM system backbone will not only ensure recycling and replenishment of parts thus promoting circular economy but will also serve to build a more productive return on investment that will strengthen the overall industry business economy.

4.1 Environmental Impact of Product Manufacturing

Manufacturing processes need raw materials, power for operating machinery and production lines, human energy, the use of coolants/lubricants etc. These are drawn from nature and consumed at a rate commensurate with the process in question. In order to assess the circularity of a product manufacturing unit, an attempt must be made to identify all such relevant parameters and quantify them. The quantification of each of these parameters is to be done first by identifying the amount of usage and converting this into a metric that describes the impact of this parameter on the environment. For example, the operation of a computer numerically controlled (CNC) machine for one hour requires power input measured in kilowatt and the usage of coolant/lubricant oils in litres. The machine operation would also require operator time measured in man-hours. In addition, the quantity of raw material used and the quantity of scrap produced in kg will also have to be measured. Each of these parameters, though measured in different units, will then have to be converted into the degree of environmental impact (scaled according to the type of resource) that will serve as a common metric. This metric is defined herein as the environmental cost (*E.Cost*) incurred in the use of that resource. The *E.Cost* is the impact on the environment as a result of the use or extraction of a particular resource (such as CNC machine, mild steel rods, cotton waste, coolant oil etc.). The *E.Cost* for each resource multiplied by the time or quantity of use will generate the environmental impact quotient (EIQ) for that particular process. The sum of the EIQ for all the processes related to a particular product will yield the circular economy quotient (CEQ) for that particular product. This is described conceptually in figure 3.

Figure 3. Impact on Environment, E.Cost and EIQ



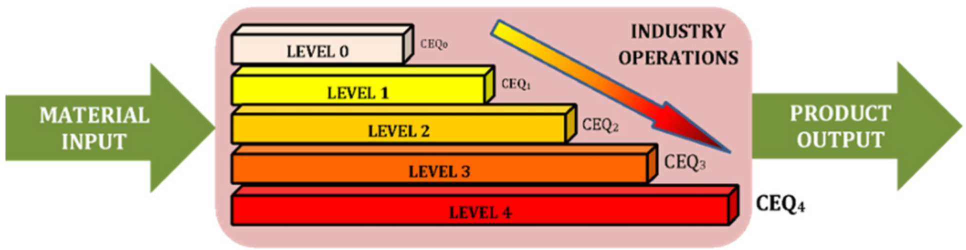
For every product that is built, the impact on the environment can be evaluated and compared through its circular economy quotient (CEQ). The CEQ is the metric to evaluate the performance of the product or its 'fit' in the circular economy. The larger the CEQ value, the greater is the impact of the product on the environment in terms of the resources it takes up. The CEQ can be used by the industry to carry out a deeper analysis of the manufacturing processes involved to minimise the impact on environment and thus serve as a quick-check method to move the specific product closer to circular economy.

4.2 Computation Of CEQ

A company starting on this journey begins at Level 0. Level 0 does not take into account the *E.Cost* of processing the raw materials, coolants, lubricants etc. that are used in manufacturing the product. The next level includes the circular economy evaluation of the inputs to the specific product along with the evaluation of its circular economic fit once it is released into the market. This is Level 1. This can be further extended to include all products of the industry and their CEQ at Level 2, all the products and the impact of their inputs at Level 3 and finally the entire operations of the industry including the supply chain logistics at Level 4. Figure 4 illustrates the CEQ waterfall diagram and the increasing level of environmental impact assessment in the form of CEQ. An industry could be evaluated for its performance as a circular economy at 5 levels.

As CE complexity increases through the levels of definition the industry's operations would need to be handled by a system with specific algorithms with high computation capacity.

Figure 4. The CEQ waterfall diagram



As a first step towards a comprehensive CE ontology, the CEQ needs to be determined for all activities that go into the operation of an industry, at each level sequentially. Companies start the CEQ at Level 0 (CEQ_0). The generic form of this expression is given by equation 1. CEQ_0 comprises six terms that, put together, describe the environmental impact quotient (EIQ) of manufacturing the product, reworking/repairing the product, remaking the product and recycling the product. The EIQ is computed based on individual environmental costs (*E.Cost*) as described earlier. The computation of CEQ at Level 0 (CEQ_0) can be described mathematically by equation 2.

$$CEQ_0 = \ddot{Q}_{PR} + \ddot{Q}_{RW} + \ddot{Q}_{RM} + \ddot{Q}_{RC} + \ddot{Q}_{RJ} - \ddot{Q}_{RP} \tag{1}$$

$$CEQ_0 = \ddot{Q}_{\left[\sum_{i=1}^{p}\left\{P_i\left(\sum_{j=1}^{q}\left(c_{mj}t_{mj}+c_{oj}t_{oj}\right)+c_r+c_h t_h\right)\right\}+C_{a_m}T_{a_m}+C_{a_o}T_{a_o}+C_{a_h}\right]}$$

$$+\ddot{Q}_{RW} + \ddot{Q}_{RM} + \ddot{Q}_{RC} + \ddot{Q}_{RJ} - \ddot{Q}_{RP} \tag{2}$$

where:

$\ddot{Q}$ Environmental Impact Quotient (EIQ)

$\ddot{Q}_{PR}$ EIQ of Manufacturing the Product

$\ddot{Q}_{RW}$ EIQ of Rework

$\ddot{Q}_{RM}$ EIQ of Remake

$\ddot{Q}_{RC}$ EIQ of Recycle

$\ddot{Q}_{RP}$ EIQ of Replenish (the usable material recovered if assumed to be made in-house)

$\ddot{Q}_{RJ}$ EIQ of Reject

P E.Cost of manufacturing each part

C_{am} E.Cost of assembly - machine

T_{am} Time of assembly – machine

C_{ao} E.Cost of assembly - operator

T_{ao} Time of assembly – operator

C_h E.Cost of material handling in assembly

p Number of parts per product

q Number of operations per part

c_m E.Cost of manufacturing (equipment usage)

c_o E.Cost of operator (during manufacturing)

t_m Time of manufacturing

t_o Time of operator

c_r E.Cost of raw material

c_h E.Cost of material handling during manufacturing

t_h Time of material handling during manufacturing

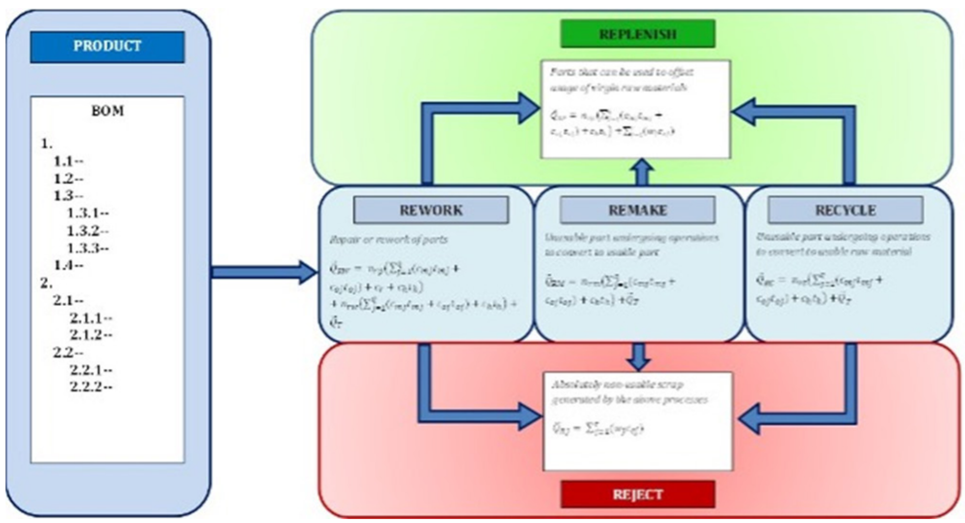
However, as part complexity and product complexity increase, computation of the EIQ just for the manufacturing operations itself becomes a very tedious task not to mention the computation of the CEQ. As mentioned earlier, this cannot be handled manually and would need a structured, database driven, computerised approach, which is described later in the paper.

4.3 Product-Based Circular Economy Implementation Framework

A simplified circular economy conceptual framework is presented as the product-based circular economy implementation (PCEI) framework. This provides a product-based industry with tools to evaluate itself on its adoption of circular economy. For a given product, the bill of material (BOM) describes the elemental breakup of the product, each element of which (either manufactured in-house or bought-out) adds its own contribution to the 'take-make-waste' path. The PCEI framework is developed as a tool to help evaluate the CEQ.

The product-based circular economy implementation (PCEIF) illustrated below in figure 5 presents a simplified approach to quantify the EIQ for the 3Rs namely 'rework', 'remake', 'recycle' and also estimate the same for the 'replenish' and 'reject' aspects of the framework. Although involving in themselves several aspects of manufacturing (similar to that described by $\ddot{Q}_{PR}$), the framework presents a simplified approach by estimating the *E.Cost* of different the different resources. Figure 5 describes the ontology of the PCEI framework through the definition of metrics that will be described in detail in this section. A total of 5 key elements form part of the framework namely ‘rework’, ‘remake’, ‘recycle’, ‘replenish’ and ‘reject’. Of these, 3 elements namely ‘rework’, ‘remake’ and ‘recycle’ form the central core of the model, or the 3Rs, to which the product BOM forms the input. Each of these 3Rs in turn either output to ‘reject’ or ‘replenish’ which, put together, will define the CEQ.

Figure 5. the EIQ for the 3Rs



4.3.1 Ontology of PCEI.Rework ($\ddot{Q}_{RW}$)

Once a product has is in service, there will be several instances when the product and/or its parts would need to be repaired or reworked to maintain its performance efficiency. Table 1 describes the semantics of this ontology element. Each of these actions amounts to a cost to the environment, the *E.Cost* of repair/rework. However, in the course of repair/rework, some of the replaced parts may be remade or recycled. These parts are not factored into the ontology of rework and will be transferred to the total stock of items remade or recycled.

Table 1. Ontology of PCEI.Rework

Element	REWORK
Definition	Any repair or rework of any element of the product that may involve either additional operations performed on the part/product and/or replacement of damaged/faulty parts
Sub-elements	- *E.Cost* of Number of parts remade - *E.Cost* of Number of parts reworked - *E.Cost* of Transportation
Computation	$\ddot{Q}_{RW} = n_{rp}\left(\sum_{j=1}^{q}\left(c_{mj}t_{mj} + c_{oj}t_{oj}\right) + c_r + c_h t_h\right) + n_{rw}\left(\sum_{j=1}^{q}\left(c_{mj}t_{mj} + c_{oj}t_{oj}\right) + c_h t_h\right) + \ddot{Q}_T$ where n_{rp} is the number of replaced parts, n_{rw} is the number of reworked parts and $\ddot{Q}_T$ is the *E.Cost* of transportation for rework $\ddot{Q}_T = \frac{d}{\eta_v}\ddot{Q}_f$ where d is distance transported in 'km', η_v is the mileage of the transport vehicle in 'km/l' and $\ddot{Q}_f$ is the exploration *E.Cost* of the fuel

4.3.2 Ontology of PCEI.Remake ($\ddot{Q}_{RM}$)

Several parts of the product will make it to 'remake' in the course of the lifecycle of the product either after having gone through rework cycles or directly pulled out of the product without rework. All these parts need to be agglomerated and processed for remake in order to extract useful economic value. Once remade, it becomes a new part that has an accrued economic value and would move into the part replenish inventory while also generating some scrap in the process that would accrue into the reject inventory. Table 2 describes the semantics of this ontology element. Each of these actions amounts to a cost to the environment, the *E.Cost* of remake.

Table 2. Ontology of PCEI.Remake

Element	REMAKE
Definition	Any part that undergoes a set of processing operations that converts the otherwise unusable part into another usable part
Sub-elements	- *E.Cost* of Number of parts remade - *E.Cost* of Transportation
Computation	$\ddot{Q}_{RM} = n_{rm}\left(\sum_{j=1}^{q}\left(c_{mj}t_{mj} + c_{oj}t_{oj}\right) + c_h t_h\right) + \ddot{Q}_T$ where n_{rm} is the number of remade parts and $\ddot{Q}_T$ is the *E.Cost* of transportation for remake $\ddot{Q}_T = \frac{d}{\eta_v}\ddot{Q}_f$ where d is distance transported in 'km', η_v is the mileage of the transport vehicle in 'km/l' and $\ddot{Q}_f$ is the exploration *E.Cost* of the fuel

4.3.3 Ontology of PCEI.Recycle ($\ddot{Q}_{RC}$)

Parts that cannot be reworked or remade will find their way into 'recycle'. These parts will require additional processing to extract base elements that in turn will serve as raw material for the manufacturing cycle. The objective is to extract the last possible value from the part before its end of life. The non-recoverable portion finds its way into the reject inventory as scrap that has its impact on the environment. Table 3 describes the semantics of this ontology element. Each of these actions amounts to a cost to the environment, the *E.Cost* of recycle.

Table 3. Ontology of PCEI.Recycle

Element	RECYCLE
Definition	Any part that undergoes a set of processing operations that converts the otherwise unusable part into a usable raw material
Sub-elements	- *E.Cost* of Number of parts recycled - *E.Cost* of Transportation
Computation	$\ddot{Q}_{RC} = n_{rc}\left(\sum_{j=1}^{q}\left(c_{mj}t_{mj} + c_{oj}t_{oj}\right) + c_h t_h\right) + \ddot{Q}_T$ where n_{rc} is the number of recycled parts and $\ddot{Q}_T$ is the *E.Cost* of transportation for recycle $\ddot{Q}_T = \frac{d}{\eta_v}\ddot{Q}_f$ where d is distance transported in 'km', η_v is the mileage of the transport vehicle in 'km/l' and $\ddot{Q}_f$ is the exploration *E.Cost* of the fuel

4.3.4 Ontology of PCEI.Replenish ($\ddot{Q}_{RP}$)

'Replenish' is the repository of all parts or usable elements that have been sourced from remake and recycle. While the above three elements contribute to a positive CEQ value, the *E.Cost* of items under 'replenish' will be deducted from the overall CEQ value. However, the computation of the EIQ of the accrued material/parts under 'replenish' is done assuming that the same has been purchased/processed by the industry. The larger the *E.Cost* of 'replenish', the more it serves to minimise the overall CEQ of the industry. Table 4 describes the semantics of this ontology element.

Table 4. Ontology of PCEI.Replenish

Element	REPLENISH
Definition	All parts or materials that are part of the product that can be converted into usable elements and from which economic value can be recovered
Sub-elements	- *E.Cost* of Number of parts produced - *E.Cost* of Quantity of material recovered
Computation	$\ddot{Q}_{RP} = n_{rp}\left(\sum_{j=1}^{q}\left(c_{mj}t_{mj} + c_{oj}t_{oj}\right) + c_h t_h\right) + \sum_{j=1}^{r}\left(w_j c_{ej}\right)$ where n_{rp} is the number or replenished parts, r is the number of different raw material types recovered, w is the weight of each raw material type recovered and c_e is the extraction cost of each raw material type

4.3.5 Ontology of PCEI.Reject ($\ddot{Q}_{RJ}$)

The cumulative effect of scrap items generated through rework, remake and recycle as well as parts that cannot be reworked, remade or recycled will amount to the total load under 'reject'. All such elements pose an individual and collective risk to the environment and cannot be further broken down into useful elements. These usually end up as landfills or in incinerators that directly have an adverse impact on environment. All industries should strive to keep this to the barest minimum as it will directly reflect on the CEQ of the industry. Table 5 describes the semantics of this ontology element.

Table 5. Ontology of Reject

Element	REJECT
Definition	All material that pose a direct risk to the environment on account of its non-biodegradability or inability to be recovered into usable elements
Sub-elements	- *E.Cost* of Non-biodegradable and Unusable Scrap generated from manufacturing, rework, remake and recycle
Computation	$\ddot{Q}_{RJ} = \sum_{j=1}^{r}\left(w_j c_{ej}\right)$ where r is the number of different material types/combinations scrapped, w is the weight of each material type/combination scrapped and c_{ej} is the extraction cost of each material type/combination Ideally: CE Compliance $\equiv \lim_{r\to 0}\sum_{j=1}^{r}\left(w_j c_{ej}\right) => \ddot{Q}_{RJ} \to 0$

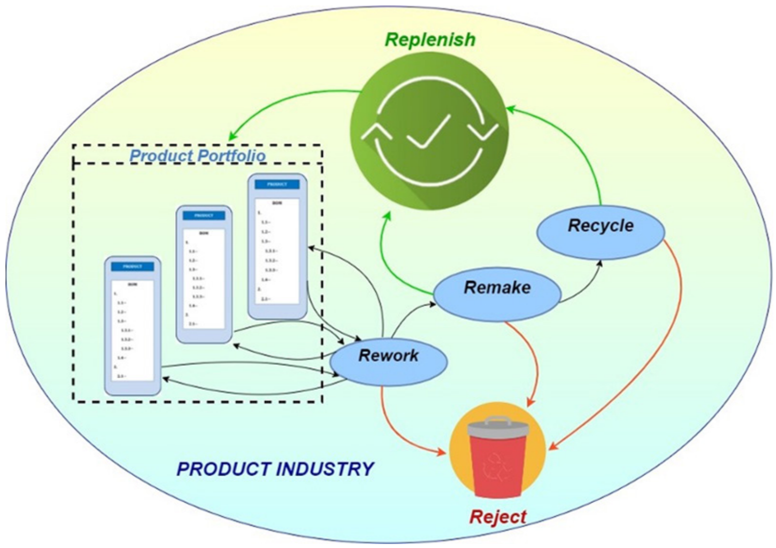
4.3.6 Obtaining CEQ from PCEI Framework

Equation 2 will provide the CEQ value of a particular product at level 0. For an industry to be operating effectively under circular economy the value of CEQ_0 should be as low as possible. This would indicate that the industry is able to effectively leverage the 3Rs and minimise its environmental impact. This can certainly be used as a scorecard to evaluate the circular economic fit of that particular product and be used as a quick litmus test for certification of a product as being CE compliant. However, this is only a minimalistic assessment of the efficacy of operation of the industry at least at an individual product level. The next section presents the approach that can be taken for assessing the industry for its circular economy compliance by evaluating the CEQ at level 2 (considering all the products of a given industry and their impact on environment).

5. OVERALL INDUSTRY CEQ EVALUATION

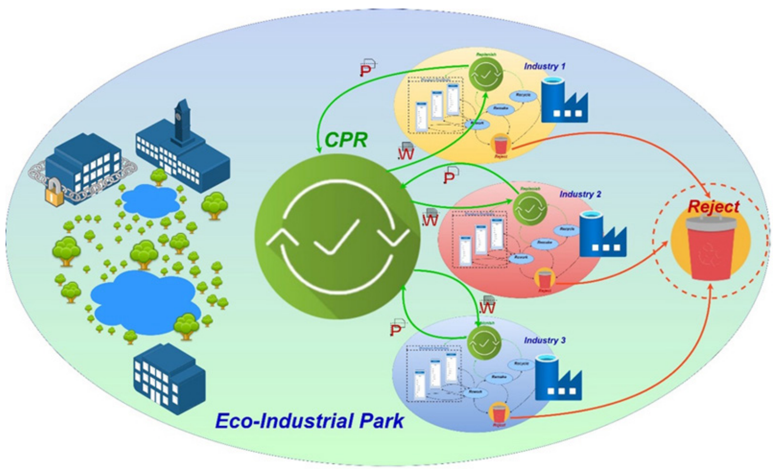
For a product with multiple components or parts, a separate CEQ evaluation needs to be carried out and for each of the parts that comprise the bill of material (BOM) of the product. In evaluating the full impact of the industry across all the product variants, an assessments is needed for CEQ values. Each product has its own BOM and therefore a PCEI scorecard needs to be developed for the entire product portfolio based on the product variety and manufacturing model (batch production or mixed-mode production). As illustrated in figure 6, all products of the industry, are represented by their respective BOMs. Each of the parts involved would go through the 3R cycle as shown in the figure. The replenish inventory serves as a supply unit to the entire portfolio with the aim of the industry being to maximise withdrawal of components from the replenish inventory for product manufacturing. The evaluation of the overall CEQ (considering the entire product portfolio) will define the overall environmental impact of the industry (CEQ_2). As in the earlier approach, CEQ_2 obtained does not consider the environmental impact of materials input to the manufacturing of the product. This framework makes it easy to get a quick assessment of the overall impact that the industry has on the environment across all its product variants.

Figure 6. Respective BOMS of each product



The creation of a central repository for closed-loop part utilization is a major step that can promote industrial symbiosis as well as reduce the demand for new raw material for industries located in a product manufacturing eco-industrial park (PMEIP). A central part repository can be created that could serve as a common inventory of parts accessible to all industries in the PMEIP as shown in figure 7. Similar to the concept of kanban in just-in-time manufacturing, an eco-production kanban (indicated as 'P' in the figure) can be generated for each batch of parts delivered by the industry to the central part repository. Similarly, an eco-withdrawal kanban (indicated as 'W' in the figure) can be generated for all batches of parts withdrawn from the central part repository by each industry. The goal of the PMEIP should be to maximise the quantity transacted through the eco-production and eco-withdrawal kanbans. This would serve to minimise new raw material usage among the member industries in the PMEIP as well as minimise the quantity of material rejected.

Figure 7. Product Manufacturing Eco-Industrial Park



Given the complexity of the estimation of the CEQ and tracking/monitoring of material going through the 3R process, a manual approach to the evaluation of the CEQ or monitoring of transactions through the central part repository for an industry or a PMEIP is too laborious. PLM platforms are widely implemented by product manufacturing companies to efficiently manage all product data. As such, PLM platforms serve as a central repository for product architecture, BOM, workflows and maintain data across the lifecycle of the product. CE can be effectively implemented in an industry by integrating the PCEI framework presented in this chapter into the PLM platform.

The BOM for a typical product as seen on a PLM platform is shown in figure 8. For each part in the BOM, there is a process associated with it and each of these processes has an impact on the environment. This impact, as explained earlier, is the CEQ that is computed for each part and updated periodically as and when it goes through rework, remake or recycle processes. Figure 8 shows several relevant columns describing data that is saved on the PLM platform for each part in the BOM. Since the PLM platform is a repository of all product data, the computation of the CEQ for each part can also be integrated into the PLM platform as shown in figure 9. While the PLM platform has several standardized data columns, a new column entitled CEQ is created into the PLM database as shown in figure 9. This column will populate the CEQ values, computed using the PCEI framework for all processes related to a given part in the BOM. The overall CEQ for the product can also be integrated into the PLM platform as shown in figure 10. The CEQ values obtained for each part can be populated in the PLM database again using the PCEI framework to yield the overall CEQ for the product as shown in figure 10. How this can be implemented in PLM has been demonstrated through the inclusion of the CEQ metadata in the data structure.

Figure 8. Typical BOM for a Product in PLM

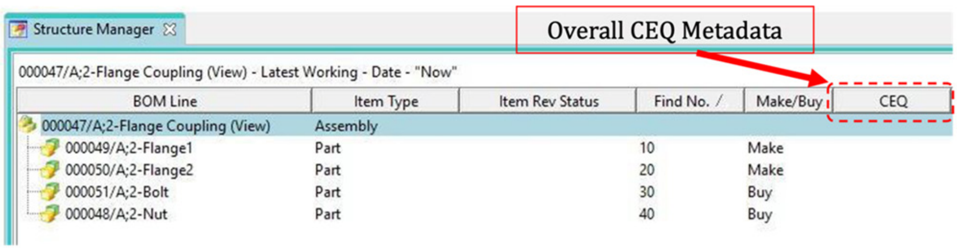


Figure 9. CEQ Implementation in PLM for a Part

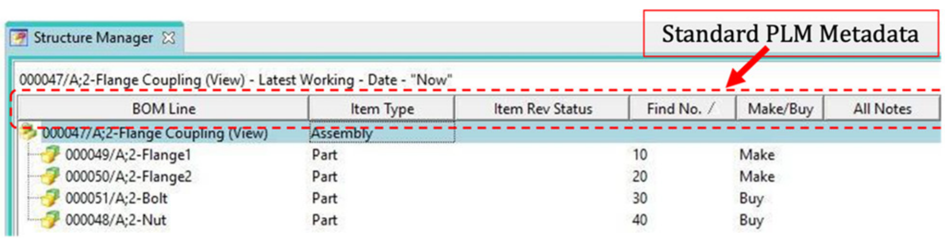
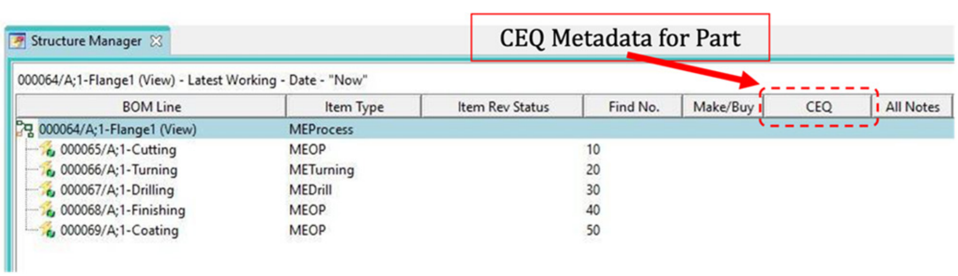


Figure 10. CEQ Implementation in PLM for a Product



The benefit of this entire approach is that it offers a product-based manufacturing industry a comprehensive system for computation of the circular economy quotient of a product by leveraging the extensive product data available in a PLM system database across its lifecycle. The use of a PLM platform enables tracking all products and their parts across their respective lifecycles thereby making it easy to estimate the impact of the product on the environment and thus the circular economy compliance of the industry.

6. DISCUSSION

The exploration of PCEI highlights a transformative approach toward sustainable manufacturing. By integrating the CEQ in the PCEI framework, this study presents a methodological advancement for quantifying and enhancing circularity in manufacturing processes, fundamentally addressing the environmental and economic challenges of linear production models. CEQ introduced in this research offers a multi-level approach to evaluating circularity within manufacturing industries, extending through five distinct levels from CEQ_0 to CEQ_4, each addressing a progressively broader scope of circular economy considerations.

Comparing the CEQ metric with existing circularity metrics like the Material Circularity Indicator (MCI) developed by Ellen MacArthur Foundation underlines CEQ's more detailed and comprehensive approach. While the MCI provides a focused measure of material reuse and recycling within products, it does not account for the broader spectrum of circular economy practices that the CEQ addresses. The CEQ's multi-level assessment, from CEQ_0 through CEQ_4, offers a deeper dive into the circular economy, evaluating not just the product and its materials but also incorporating the environmental impacts of production inputs, all products of an industry, and the comprehensive operations including supply chain logistics. This depth of analysis allows for a nuanced understanding of an industry's circular economy performance that the MCI, with its narrower focus, might not fully capture. The CEQ metric, by offering a granular and holistic view of circularity across multiple dimensions of an industry's operations, provides a more robust tool for assessing and driving improvements in circular economy practices. This comparison underscores the need for diverse tools in the circular economy toolkit, each contributing unique insights into the complex landscape of sustainable manufacturing.

However, this comprehensive approach also introduces significant challenges, particularly in data collection and analysis at higher levels like CEQ_3 and CEQ_4. These challenges highlight the need for expanding the definition of the current CEQ metric, necessitating further research into developing more efficient computational tools and algorithms to facilitate data management and analysis, thus making the CEQ metric more accessible and applicable across various industry settings.

The framework's practicality echoes the increasing discourse on circular economy's role in fostering sustainable manufacturing, highlighting the essentiality of technological adoption in achieving circularity. Furthermore, the framework's application through PLM systems illustrates a tangible response to calls for integrating circular economy principles into the fabric of organizational practices, thereby advancing the agenda of sustainable development within the manufacturing sector.

PCEI underscores five key elements: rework, remake, recycle, replenish, and reject. Each element is meticulously designed to quantify the environmental cost (E.Cost) associated with a product's lifecycle stages, from production through to end-of-life. This framework's ontology, including its focus on the 3Rs as the core model components, positions it as a pragmatic tool for assessing and improving product circularity within industries. For instance, the ontology of rework within PCEI provides a formulaic approach to quantifying environmental costs associated with product maintenance, a specificity absent in broader frameworks.

However, the PCEI framework's detailed and structured approach may also introduce limitations. One such limitation is its potential complexity and resource intensity, which could pose challenges for small and medium-sized enterprises (SMEs) lacking the necessary technological infrastructure or expertise to implement the framework fully. This contrasts with more flexible frameworks that might offer simpler, less resource-intensive entry points into circular economy practices. Moreover, the reliance on advanced technologies like PLM systems, while advantageous for tracking and managing product lifecycles, might not be feasible for all organizations, particularly those in developing countries or sectors with lower technological adoption rates. This underscores the need for product manufacturing industries to develop lighter data management solutions ensuring a broader adoption of CE principles.

The limitations of the PCEI framework highlights the necessity for future research to explore more accessible and scalable circular economy implementation strategies. Future studies could focus on developing modular or scalable versions of the PCEI framework that can be tailored to the capacities and resources of different types of organizations. Additionally, research could investigate the integration of alternative technologies that are more readily available to SMEs, thereby broadening the framework's applicability and facilitating a more inclusive transition towards circular economy practices across industries and regions.

In conclusion, this paper encapsulates a comprehensive approach to embedding circular economy principles within the manufacturing sector, addressing both the quantitative and qualitative aspects of circularity. The development and integration of CEQ and PCEI not only respond to the academic and industrial call for actionable circular economy frameworks but also pave the way for future explorations into sustainable manufacturing practices. This work, firmly rooted in the literature, extends the discussion on circular economy beyond theory, offering practical insights and tools for industry practitioners and policymakers alike. In summary the broader implication of this research is represented below:

Table 6. Implications of the research

Category	Description
Environmental	The CPR business model in conjunction with the PMEIP as described as a part of this research highlights the environmental benefits of implementing CE in manufacturing, such as significant reductions in waste generation, improved resource efficiency, and lower carbon footprint. This aligns with global sustainability goals and addresses pressing environmental issues Geissdorfer et. al (2017).
Economic	The effective implementation of CPR and PMEIP ensures the economic implications, including the potential for cost savings through efficient resource use, the creation of new business opportunities in the recycling and remanufacturing sectors, and the long-term financial benefits of sustainable manufacturing practices (Ghisellini, Cialini & Ulgiati, 2016). PMEIP model which utilizes the circular part repository to repurpose the end-of-life products can realize its maximum potential by eliminating the costs involved in the supply chain of sourcing.
Technological	This research takes a huge technological leap in establishing a working relationship between the developed framework and the PLM system. Addresses the need for and challenges of integrating advanced technologies like AI, IoT, and PLM in manufacturing processes. This includes the potential for improved efficiency, but also the challenges of upskilling the workforce and investing in new technology (Carvalho et al, 2018).
Educational	Suggests the importance of educational initiatives aimed at increasing awareness and understanding of CE principles among all stakeholders, including manufacturers, consumers, and policymakers. This could involve training programs, academic courses, and public awareness campaigns (Stark, 2020).

7. FUTURE RESEARCH

In the pursuit of quantification of the circularity of an industry, this pilot study established a working model of the PCEI framework and its integration with the PLM system, to come up with a score for a single product at level 0 of the described CEQ metric. This leaves room for the future research to explore the possibility of validating how the framework functions at various levels of CEQ quantification.

Empirical validation of this framework in diverse manufacturing contexts at various levels, remains a key area for future investigation which also encompasses assessing its applicability and effectiveness across various industries and geographical locations.

The exploration of technological innovations, especially the integration of PLM systems with emerging technologies like AI and blockchain, presents a significant avenue for advancing circular economy practices Bocken, Nuzum, Weissbrod (2021). Equally important is the need for comprehensive economic analyses and consumer behaviour studies to understand the broader impact of circular economy implementation. These future research directions as indicated in Table 7 not only build upon the current findings but also set the stage for the ensuing recommendations, ensuring that they are informed by a forward-looking perspective and a keen understanding of the evolving landscape in sustainable manufacturing.

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